Z10.5
(cont)

TABLE IV / Sheet 2

REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT THOSE AREAS OF SUBSTANTIAL CORROSION OF DOUBLE SKIN BULK CARRIERS WITHIN THE CARGO LENGTH AREA

DECK STRUCTURE INCLUDING CROSS STRIPS, MAIN CARGO HATCHWAYS, HATCH COVERS, COAMINGS AND TOPSIDE TANKS		
Structural member	Extent of measurement	Pattern of measurement
Cross Deck Strip plating	Suspect Cross Deck Strip plating	Five-point pattern between underdeck siffeners over 1 metre length
Underdeck Stiffeners	Transverse members Longitudinal member	Five-point pattern at each end and mid span Five-point pattern on both web and flange
Hatch Covers	Side and end skirts, each three locations Three longitudinal bands, outboard strakes (2) and centreline strake (1)	Five-point pattern at each location Five-point measurement each band
Hatch Coamings	Each side and end of coaming, one band lower 1/3, one band upper 2/3 of coaming	Five-point measurement each band i.e. end or side coaming
Topside Ballast Tanks	a) watertight transverse bulkheads: - Lower 1/3 of bulkhead - Upper 2/3 of bulkhead - Stiffeners	 Five-point pattern over 1 sq. metre of plating Five-point pattern over 1 sq. metre of plating Five-point pattern over 1 metre length
Topside Ballast Tanks	b) two representative swash transverse bulkheads: - Lower 1/3 of bulkhead - Upper 2/3 of bulkhead - Stiffeners	 Five-point pattern over 1 sq. metre of plating Five-point pattern over 1 sq. metre of plating Five-point pattern over 1 metre length
Topside Ballast Tanks	c) three representative bays of slope plating - Lower 1/3 of tank - Upper 2/3 of tank	 Five-point pattern over 1 sq. metre of plating Five-point pattern over 1 sq. metre of plating
Topside Ballast Tanks	d) Longitudinals, suspect and adjacent	Five-point pattern on both web and flange over 1 metre length
Main Deck Plating	Suspect plates and adjacent (4)	Five-point pattern over 1 sq. metre of plating
Main Deck Longitudinals	Suspect Plates	Five-point pattern on both web and flange over 1 metre length
Web Frames/Transverses	Suspect Plates	Five-point pattern over 1 sq. metre

Z10.5
(cont)

TABLE IV / Sheet 3

REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT THOSE AREAS OF SUBSTANTIAL CORROSION OF DOUBLE SKIN BULK CARRIERS WITHIN THE CARGO LENGTH AREA

STRUCTURE IN DOUBLE SIDE SPACES OF DOUBLE SKIN BULK CARRIERS INCLUDING WING VOID SPACES OF ORE CARRIERS		
Structural member	Extent of measurement	Pattern of measurement
Side shell and inner plating: - Upper strake and strakes in way of horizontal girders - All other strakes	 - Plating between each pair of transverse frames / longitudinals in a minimum of three bays (along the tank) - Plating between every third pair of longitudinals in same three bays	 - Single measurement - Single measurement
Side shell and inner side transverse frames / longitudinals on: - upper strake - all other strakes	 - Each transverse frame / longitudinal in same three bays - Every third transverse frame / longitudinal in same three bays	 - Three measurements across web and 1 measurement on flange - Three measurements across web and 1 measurement on flange
Transverse frames / Longitudinals - brackets	Minimum of three at top, middle and bottom of tank in same three bays	Five-point pattern over area of bracket
Vertical web and transverse bulkheads: - strakes in way of horizontal girders - other strakes	 - Minimum of two webs and both transverse bulkeads - Minimum of two webs and both transverse bulkeads	 - Five-point pattern over approx. two square metre area - Two measurements between each pair of vertical stiffeners
Horizontal girders	Plating on each girder in a minimum of three bays	Two measurements between each pair of longitudinal girder stiffeners
Plate stiffening	Where applicable	Single measurements

Z10.5
(cont)

TABLE IV / Sheet 4

REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT THOSE AREAS OF SUBSTANTIAL CORROSION OF DOUBLE SKIN BULK CARRIERS WITHIN THE CARGO LENGTH AREA

TRANSVERSE BULKHEADS IN CARGO HOLDS		
Structural member	Extent of measurement	Pattern of measurement
Lower stool, where fitted	- Transverse band within 25mm of welded connection to inner bottom - Transverse bands within 25mm of welded connection to shelf plate	- Five-point pattern between stiffeners over one metre length - Five-point pattern between stiffeners over one metre length
Transverse bulkheads	- Transverse band at approximately mid height - Transverse band at part of bulkhead adjacent to upper deck or below upper stool shelf plate (for those ships fitted with upper stools)	- Five-point pattern over one square metre of plating - Five-point pattern over one square metre of plating

Z10.5
(cont)

TABLE V

MINIMUM REQUIREMENTS OF OVERALL AND CLOSE-UP SURVEY AND THICKNESS MEASUREMENTS AT INTERMEDIATE SURVEY OF DOUBLE SKIN BULK CARRIERS

Age of ship at time of intermediate survey due date		
5 < age ≤ 10	10 < age ≤ 15	age > 15
Overall survey of Representative ballast tanks selected by the attending surveyor (the selection is to include fore and aft peak tanks and a number of other tanks, taking into account the total number and type of ballast tanks)	The requirements of the previous special survey (see 4.2.3)	The requirements of the previous special survey (see 4.2.4)
Overall and close-up survey of Suspect Areas identified at previous surveys		
Overall survey of all cargo holds		
Thickness measurements to an extent sufficient to determine both general and local corrosion levels at areas subject to close-up survey at 'suspect areas' identified at previous surveys		

Z10.5
(cont)

TABLE VI

PROCEDURES FOR CERTIFICATION OF FIRMS ENGAGED IN THICKNESS MEASUREMENT OF HULL STRUCTURES

1. Application

This guidance applies for certification of the firms which intend to engage in the thickness measurement of hull structures of the vessels.

2. Procedures for Certification

(1) Submission of Documents:

Following documents are to be submitted to the Society for approval:

a) Outline of firm, e.g. organization and management structure.

b) Experience of the firm on thickness measurement inter alia of hull structures of the vessels.

c) Technicians' careers, i.e. experience of technicians as thickness measurement operators, technical knowledge of hull structure, etc. Operators, are to be qualified according to a recognized industrial NDT Standard.

d) Equipment used for thickness measurement such as ultra-sonic testing machines and its maintenance/calibration procedures.

e) A guide for thickness measurement operators.

f) Training programmes of technicians for thickness measurement.

g) Measurement record format in accordance with the Recommended Procedures for Thickness Measurements of Bulk Carriers, contained in Annex II.

(2) Auditing of the firm:

Upon reviewing the documents submitted with satisfactory results, the firm is audited in order to ascertain that the firm is duly organised and managed in accordance with the documents submitted, and eventually is capable of conducting thickness measurement of the hull construction of the ships.

(3) Certification is conditional on an onboard demonstration at thickness measurements as well as satisfactory reporting.

3. Certification

(1) Upon satisfactory results of both the audit of the firm in 2(2) and the demonstration tests in 2(3) above, the Society will issue a Certificate of Approval as well as a notice to the effect that the thickness measurement operation system of the firm has been certified by the Society.

(2) Renewal/endorsement of the Certificate is to be made at intervals not exceeding 3 years by verification that original conditions are maintained.

Z10.5
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4. Information of any alteration to the Certified Thickness Measurement Operation System

In case where any alteration to the certified thickness measurement operation system of the firm is made, such an alteration is to be immediately informed to the Society. Re-audit is made where deemed necessary by the Society.

5. Cancellation of Approval

Approval may be cancelled in the following cases:

(1) Where the measurements were improperly carried out or the results were improperly reported.

(2) Where the Society's surveyor found any deficiencies in the approved thickness measurement operation systems of the firm.

(3) Where the firm failed to inform of any alteration in 4 above to the Society.

Z10.5
(cont)

TABLE VII

SURVEY REPORTING PRINCIPLES

As a principle, for bulk carriers subject to ESP, the surveyor is to include the following content in his report for survey of hull structure and piping systems, as relevant for the survey.

The structure of the reporting content may be different, depending on the report system for the respective Societies.

1. General

1.1 A survey report is to be generated in the following cases:

- In connection with commencement, continuation and / or completion of periodical hull surveys, i.e. annual, intermediate and special surveys, as relevant

- When structural damages / defects have been found
- When repairs, renewals or modifications have been carried out
- When condition of class has been imposed or deleted

1.2 The purpose of reporting is to provide:

- Evidence that prescribed surveys have been carried out in accordance with applicable classification rules
- Documentation of surveys carried out with findings, repairs carried out and condition of class imposed or deleted
- Survey records, including actions taken, which shall form an auditable documentary trail. Survey reports are to be kept in the survey report file required to be on board
- Information for planning of future surveys
- Information which may be used as input for maintenance of classification rules and instructions

1.3 When a survey is split between different survey stations, a report is to be made for each portion of the survey. A list of items surveyed, relevant findings and an indication of whether the item has been credited, are to be made available to the next attending surveyor, prior to continuing or completing the survey. Thickness measurement and tank testing carried out is also to be listed for the next surveyor.

2. Extent of the survey

2.1 Identification of compartments where an overall survey has been carried out.

2.2 Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where a close-up survey has been carried out, together with information of the means of access used.

2.3 Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where thickness measurement has been carried out.

Note: As a minimum, the identification of location of close-up survey and thickness measurement is to include a confirmation with description of individual structural members corresponding to the extent of requirements stipulated in UR Z10.2 based on type of periodical survey and the ship's age.

Z10.5 (cont)

Where only partial survey is required, i.e. 25% of shell frames, one transverse web, two selected cargo hold transverse bulkheads, the identification is to include location within each ballast tank and cargo hold by reference to frame numbers.

2.4 For areas in ballast tanks and cargo holds where protective coating is found to be in GOOD condition and the extent of close-up survey and / or thickness measurement has been specially considered, structures subject to special consideration are to be identified.

2.5 Identification of tanks subject to tank testing.

2.6 Identification of piping systems on deck and within cargo holds, ballast tanks, pipe tunnels, cofferdams and void spaces where:

- Examination including internal examination of piping with valves and fittings and thickness measurement, as relevant, has been carried out

- Operational test to working pressure has been carried out

3. Result of the survey

3.1 Type, extent and condition of protective coating in each tank, as relevant (rated GOOD, FAIR or POOR).

3.2 Structural condition of each compartment with information on the following, as relevant:

- Identification of findings, such as:
  - Corrosion with description of location, type and extent
  - Areas with substantial corrosion
  - Cracks / fractures with description of location and extent
  - Buckling with description of location and extent
  - Indents with description of location and extent
- Identification of compartments where no structural damages / defects are found

The report may be supplemented by sketches / photos.

3.3 Thickness measurement report is to be verified and signed by the surveyor controlling the measurements on board.

4. Actions taken with respect to findings

4.1 Whenever the attending surveyor is of the opinion that repairs are required, each item to be repaired is to be identified in the survey report. Whenever repairs are carried out, details of the repairs effected are to be reported by making specific reference to relevant items in the survey report.

4.2 Repairs carried out are to be reported with identification of:
- Compartment
- Structural member
- Repair method (i.e. renewal or modification) including:
  - steel grades and scantlings (if different from the original);
  - sketches/photos, as appropriate.
- Repair extent
- NDT / Tests

Z10.5 (cont)

4.3 For repairs not completed at the time of survey, condition of class is to be imposed with a specific time limit for the repairs. In order to provide correct and proper information to the surveyor attending for survey of the repairs, condition of class is to be sufficiently detailed with identification of each item to be repaired. For identification of extensive repairs, reference may be given to the survey report.

Z10.5
(cont)

TABLE VIII (i)

IACS UNIFIED REQUIREMENTS FOR ENHANCED SURVEYS
EXECUTIVE HULL SUMMARY

Issued upon Completion of Special Survey

GENERAL PARTICULARS

SHIP'S NAME:

CLASS IDENTIFY NUMBER:

IMO IDENTIFY NUMBER:

PORT OF REGISTRY:

NATIONAL FLAG:

DEADWEIGHT (M. TONNES):

GROSS TONNAGE:
NATIONAL:
ITC (69):

DATE OF BUILD:

CLASSIFICATION NOTATION:

DATE OF MAJOR CONVERSION:

TYPE OF CONVERSION:

a) The survey reports and documents listed below have been reviewed by the undersigned and found to be satisfactory

b) A summary of the survey is attached herewith on sheet 2

c) The hull special survey has been completed in accordance with the Regulations on [date]

Executive Summary Report completed by:	Name Signature	Title
OFFICE	DATE	
Executive Summary Report verified by:	Name Signature	Title
OFFICE	DATE	

Attached reports and documents:
1)
2)
3)
4)
5)
6)

Z10.5
(cont)

TABLE VIII (ii)

EXECUTIVE HULL SUMMARY

A)	General Particulars:	-	Ref.Table VIII (i)
B)	Report Review:	-	Where and how survey was done
C)	Close-up Survey:	-	Extent (Which tanks)
D)	Thickness measurements:	-	Reference to Thickness Measurement report
		-	Summary of where measured
		-	Separate form indicating the tanks/areas with Substantial Corrosion, and corresponding
		-	Thickness diminution
		-	Corrosion pattern
E)	Tank / hold / double-side skin void space Protection	-	Separate form indicating:
		-	Location of coating
		-	Condition of coating (if applicable)
F)	Repairs:	-	Identification of tanks/areas
G)	Conditions of Class:		
H)	Memoranda:	-	Acceptable defects
		-	Any points of attention for future surveys, e.g. for Suspect Areas.
		-	Examination of ballast tanks and double-side skin void spaces at annual surveys due to coating breakdown
I)	Conclusion:	-	Statement on evaluation/verification of survey report

Z10.5
(cont)

TABLE VIII (iii) A – non CSR vessels

EXTRACT OF THICKNESS MEASUREMENT

Reference is made to the thickness measurements report:

Position of substantially corroded Tanks/Areas or Areas with deep pitting[1)]	Thickness diminution[%]	Corrosion pattern[2)]	Remarks: e.g. Ref. attached sketches

Remarks:

1) Substantial corrosion, i.e. 75 – 100% of acceptable margins wasted.

2) P = Pitting; C = Corrosion in General
Any bottom plating with a pitting intensity of 20% or more, with wastage in the substantial corrosion range or having an average depth of pitting of 1/3 or more of actual plate thickness is to be noted.

Z10.5
(cont)

TABLE VIII (iii) B – CSR vessels

EXTRACT OF THICKNESS MEASUREMENTS

Reference is made to the thickness measurements report:

1) Position of substantially corroded Tanks/Areas or Areas with deep pitting	t_m - t_{ren} (mm)	2) Corrosion pattern	Remarks: e.g. Ref. Attached sketches

Remarks

1) Substantial corrosion, an extent of corrosion such that the assessment of the corrosion pattern indicates a measured thickness between t_{ren} + 0.5mm and t_{ren}.

2) P = Pitting
C = Corrosion in General
Areas with deep pitting assessed according to 8.2 are to be recorded in this column.

Z10.5
(cont)

TABLE VIII (iv)

TANK PROTECTION

Tank/hold/void nos.[1)]	Tank/hold/void protection[2)]	Coating condition[3)]	Remarks

Remarks:

1) All ballast tanks, cargo holds and double-side skin void spaces to be listed.

2) C = Coating; NP = No Protection

3) Coating condition according to the following standard:

GOOD condition with only minor spot rusting.

FAIR condition with local breakdown at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition.

POOR condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.

For ballast tanks, if coating condition less than **"GOOD"** is given, tanks are to be examined at annual surveys. This is to be noted in part H) of the Executive Hull Summary.

For double-side skin void spaces on bulk carriers exceeding 20 years of age and of 150 m in length and upwards, if coating condition POOR is given, those voids are to be examined at annual surveys. This is to be noted in part H of Executive Hull Summary.

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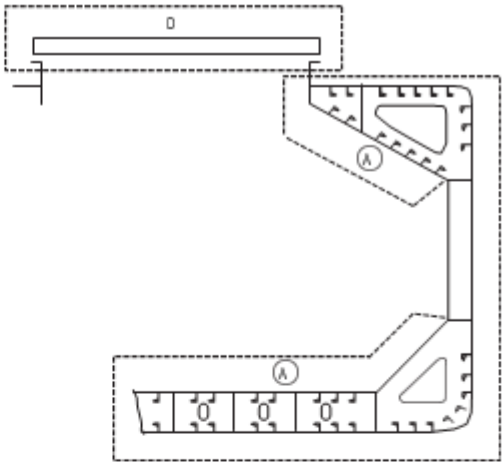
Typical transverse section

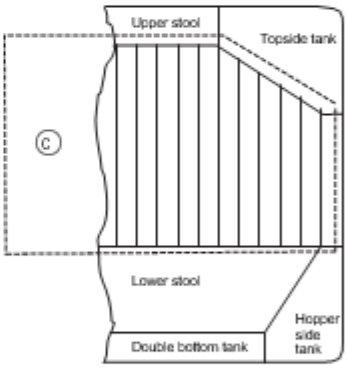
Areas (A) and (D)



Thickness to be reported on TM3-BC, TM4-BC, TM6-BC and TM7-BC as appropriate

A cargo hold, transverse bulkhead

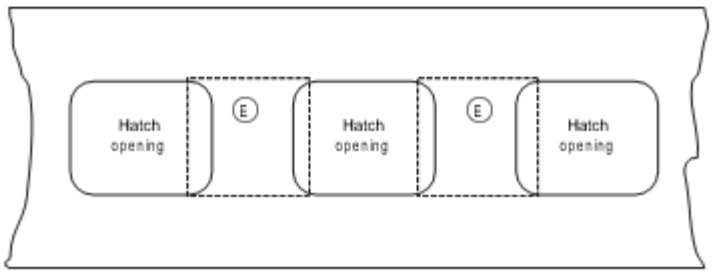
Area (C)



Thickness to be reported on TM5-BC

Typical areas of deck plating and underdeck structure inside line of hatch openings between cargo hold hatches

Area (E)



Thickness to be reported on TM6-BC

Figure 10 - Close-up survey and thickness measurement areas

Z10.5
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Area (B)

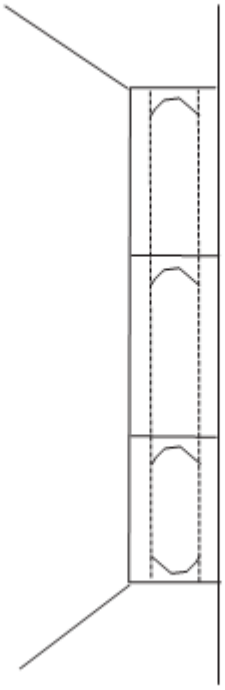

Figure 11(a) - Close-up survey and thickness measurement areas ordinary transverse frame in double skin tank

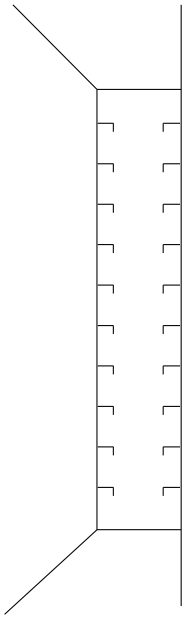

Figure 11(b) - Close-up survey and thickness measurement areas ordinary longitudinal structure in double skin tank

End of Section

Z10.5
(cont)

ANNEX I

GUIDELINES FOR TECHNICAL ASSESSMENT IN CONJUNCTION WITH PLANNING FOR ENHANCED SURVEYS OF DOUBLE SKIN BULK CARRIERS SPECIAL SURVEY– HULL

Contents:

1. INTRODUCTION

2. PURPOSE AND PRINCIPLES
2.1 Purpose
2.2 Minimum Requirements
2.3 Timing
2.4 Aspects to be considered

3. TECHNICAL ASSESSMENT

3.1 General
3.2 Methods
3.2.1 Design Details
3.2.2 Corrosion
3.2.3 Locations for Close-up Survey and Thickness Measurement

REFERENCES

1. IACS Recommendation 76, Guidelines for Surveys, Assessment and Repair of Hull Structures – Bulk Carriers, 2007
2. TSCF, "Guidelines for the Inspection and Maintenance of Double Hull Tanker Structures, 1995"
3. TSCF, "Guidance Manual for Tanker Structures, 1997"

1. INTRODUCTION

These guidelines contain information and suggestions concerning technical assessments, which may be of use in conjunction with the planning of enhanced Special Surveys of double skin bulk carriers. As indicated in 5.1.5 of IACS Unified Requirement Z10.5, "Hull Surveys of Double Skin Bulk Carriers" (Ref. 1), the guidelines are a recommended tool which may be invoked at the discretion of an IACS Member Society, when considered necessary and appropriate, in conjunction with the preparation of the required Survey Programme.

2. PURPOSE AND PRINCIPLES

2.1 Purpose

The purpose of the technical assessments described in these guidelines is to assist in identifying critical structural areas, nominating suspect areas and in focusing attention on structural elements or areas of structural elements which may be particularly susceptible to, or evidence a history of, wastage or damage. This information may be useful in nominating locations, areas holds and tanks for thickness measurement, close-up survey and tank testing.

Critical Structural Areas are locations which have been identified from calculations to require monitoring or from the service history of the subject ship or from similar or sister ships (if

Z10.5 (cont)

available) to be sensitive to cracking, buckling or corrosion which would impair the structural integrity of the ship.

2.2 Minimum Requirements

However, these guidelines may not be used to reduce the requirements pertaining to thickness measurement, close-up survey and tank testing contained in Tables I and II and in paragraph 2.5, respectively, of Z10.5, which are, in all cases, to be complied with as a minimum.

2.3 Timing

As with other aspects of survey planning, the technical assessments described in these guidelines should be worked out by the Owner or operator in cooperation with the Classification Society well in advance of the commencement of the Special Survey, i.e. prior to commencing the survey and normally at least 12 to 15 months before the survey's completion due date.

2.4 Aspects to be considered

Technical assessments, which may include quantitative or qualitative evaluation of relative risks of possible deterioration, of the following aspects of a particular ship may be used as a basis for the nomination of holds, tanks and areas for survey:

- Design features such as stress levels on various structural elements, design details and extent of use of high tensile steel.
- Former history with respect to corrosion, cracking, buckling, indents and repairs for the particular ship as well as similar vessels, where available.
- Information with respect to types of cargo carried, use of different holds/tanks for cargo/ballast, protection of holds and tanks and condition of coating, if any.

Technical assessments of the relative risks of susceptibility to damage or deterioration of various structural elements and areas are to be judged and decided on the basis of recognized principles and practices, such as may be found in publications of IACS (Ref. 1) and Tanker Structure Cooperative Forum (TSCF), (Refs. 2 and 3).

3. TECHNICAL ASSESSMENT

3.1 General

There are three basic types of possible failure, which may be the subject of technical assessment in connection with planning of surveys; **corrosion**, **cracks** and **buckling**. Contact damages are not normally covered by the survey planning since indents are usually noted in memoranda and assumed to be dealt with as a normal routine by Surveyors.

Technical assessments performed in conjunction with the survey planning process are, in principle, to be as shown schematically in Figure 1. The approach is basically an evaluation of the risk in the following aspects based on the knowledge and experience related to:
- Design
- Corrosion.

The design is to be considered with respect to structural details, which may be susceptible to buckling or cracking as a result of vibration, high stress levels or fatigue.